

B.Tech in Biomedical Engineering

Curriculum

	Course code	Course name	Category	Credits	Prereq
Sem-1	EP1108	Modern Physics	BS	2	None
	LA1760	English Communication	LA	2	None
	EP1118	Mathematics for Physics	BS	2	None
	MA1110	Calculus - I	BS	1	None
	MA1220	Calculus - II	BS	1	None
	BM1000	Physio-anatomy-I	Core	2	None
	ID1063	Introduction to Programming	BE	3	None
	BT1010	Life Science	BS	1	None
	CY1010	Environmental Chemistry	BS	2	None
	LAXX	LA elective	LA	1	None
Sem Total				17	

	Course code	Course name	Category	Credits	Prereq
Sem-2	ID1054	Digital fabrication	BE	2	None
	ME1020	Engineering Mechanics	BE	3	None
	BM1030	Bioengineering	Core	2	None
	EP1208	Electricity & magnetism	BS	2	None
	EM3020	Introduction to Entrepreneurship	LA	1	None
	ID1050	Artificial intelligence	BE	1	None
	MA1140	Elementary Linear algebra	BS	1	None
	MA1150	Differential equations	BS	1	None
	BM1010	Physio-anatomy-II	Core	2	BM1000
	LAXX	LA elective	LA	1	None
Sem Total				16	

	Course code	Course name	Category	Credits	Prereq
Sem-3	ID2230	Data structures and applications	BE	3	None
	BM2043	Algorithms and data structures lab	Core	2	
	BM2053	Mathematical models & systems biology	Core	2	BM1000; BM1010
	MA2110	Introduction to Probability	BS	1	None
	MA2120	Transform techniques	Core	1	
	BM2013	Analog and integrated circuits	Core	3	EE1100
		Free Electives	FE	3	
	LAXXX	LA Electives	LA	2	None
Sem Total				17	

	Course code	Course name	Category	Credits	Prereq
Sem-4	MA2130	Complex variables	BS	1	MA1110
	BM2000	Control systems	BE	1	None
	BM2003	Introduction to embedded systems	Core	3	BM2013
	BM2023	Basic Bioinformatics	Core	2	BM1000; BM1010
	BM3030	Medical nanotech-I	Core	2	BM1000; BM1010
	BM3090	Biomedical imaging	Core	3	None
	MA2140	Introduction to Statistics	BS	1	None
	LAXXX	LA Electives	LA	2	None
		Free Electives	FE	2	
Sem Total				17	

	Course code	Course name	Category	Credits	Prereq
Sem-5	BM3000	Natural intelligence, Foundations of		3	BM2033; BM1000; BM1010
	BM3023	Cell Technology	Core	3	BM1000; BM1010
	BM3040	Introduction to Biomechanics	Core	2	BM1000; BM1010
	BM5013	Sensors & transducers in healthcare	Core	2	BM2013; BM2003
	BM5090	Biomaterials-1	Core	2	BM1000; BM1010
	BM3900	Linear Systems and Signals	BE	3	None
		Free Electives	FE	2	
Sem Total				17	

	Course code	Course name	Category	Credits	Prereq
Sem-6		Free Electives	FE	6	required CGPA
		Dept Electives	DE	8	
Sem Total				14	

	Course code	Course name	Category	Credits	Prereq
	BM5023	Biomedical devices	Core	2	BM5013
		Dept Electives	DE	12	

	Course code	Course name	Category	Credits	Prereq
	ID4066	Ethics and Values	LA	1	
	BM4000	Regulatory processes and bioethics	Core	2	BM5023

[illegible]

		BM4040	Mechano-biology	MEC	3	BM3040		AI4000	Robotics	NI	3	
		BM4051	Computational Biomechanics Lab	MEC	2	BM3040		BM2033	Probability and Random Processes	NI	2	MA2110
		BM4060	Movement science	MEC	2	BM3000		BM4080	Medical image processing & analysis	NI	2	ID<sp>
		BM5110	Lab-on-chip	MEC	1			BM4021	Medical image processing lab	NI	2	BM4080
		BM5141	Biomaterials-2	MEC	2	BM5090		BM4025	Natural intelligence, Implementations of	NI	2	BM3000
		BM5160	Mechanics of bio-fluids	MEC	2	BM3040		BM4060	Movement science	NI	2	BM3000
		BM6070	Biomicrofluidics	MEC	2	BM5110		BM4061	Movement sciences lab	NI	2	BM4060
		BM6080	Advanced Biomechanics	MEC	2	ID;BM3040		BM5080	Neurotechnology & BCI theory	NI	2	ID<sp>;BM3000
		ME3030	Modelling and simulation	MEC	3			BM5081	Neurotechnology & BCI lab	NI	2	BM5080
		MS2050	Mechanical behavior of materials	MEC	3			BM4081	Computational neuroscience lab	NI	2	BM6140
		BM4135	Biomechanical Design	MEC	2	BM3040		BM6140	Theoretical & computational neuroscience	NI	2	BM3000
		BM5033	Statistical Inference in Bioengineering	MEC	2			CS3390	Machine learning, Foundations of	NI	3	
		BM5XX3	Bayesian Inference in Bioengineering	MEC	2	BM5033		BM4021	Medical Image Processing Lab	IN	2	
		BM5XX3	Mathematical Physiology & Systems Medicine	MEC	2			BM4080	Medical Image Processing and Analysis	IN	2	
		BM5020	AI in Biomedicine and Healthcare	MEC	3			BM5033	Statistical Inference in Bioengineering	IN	2	
			Elective bucket total		35			BM5020	AI in Biomedicine and Healthcare	IN	3	

























Course descriptions

B.Tech in Biomedical Engineering

Course code : BM1030
Course name : Bioengineering
Num of credits : 2
Semester : 2
Prerequisites : None

Course Description

This course gives a rigorous overview of the various fields of bioengineering. It covers basic concepts of Biomedical Engineering and connects the biological systems to the common theme of engineering analysis. The highly interdisciplinary nature of Bioengineering across various engineering disciplines will be elaborated in this course. This course will serve as the basis for students from interdisciplinary background to explore various topics in bioengineering.

Course contents

The topics addressed in this course include origin of signals in biosystems and living organisms, their sensing, detection and meaningful processing for practical diagnostic sensing applications. Various engineering aspects of the detection, acquisition, processing, and display of signals etc will be addressed in this course. The students will get exposure to the research activities that are being carried out in the department.

References

1. John G. Webster-Medical Instrumentation Application and Design, 4th Edition (2009)

#REF!

Course code : BM4015
Course name : Capstone Project
Num of credits : 3
Semester : 8
Prerequisites : Completion of 15 credits in Elective basket

Course Description

This is a project based on the elective basket chosen.

Course contents

References

Course code : BM4010
Course name : Medical nanotech-II
Num of credits : 2
Semester : Even
Prerequisites : BM3030: Medical nanotech-I

Course Description

This course deals with the medical applications of Nanoscience and Nanotechnology. This is a Theory & lab course. Students will be exposed to various nanosynthesis methodologies. There will be "hands -on" training involving synthesis and characterization of nanoparticles (to be used as nanomedicines).

Course contents

Synthesis of different nanomaterials system for nanomedicine applications. Such as polymer nanoparticles, metal nanoparticles, lipid nanoparticles, oxide nanoparticles, biofunctionalized nanoparticles, Nanofibers, Nono-Rod.

References

1. Nanotechnology in Biology and Medicine: Methods, devices, and applications. Ed. Tuan Vo-Dinh. Boca Raton, FL: CRC Press, Taylor & Francis Group, 2007. ISBN-13: 978-0-8493-2949-4
2. Relavant Journal Publications

Course code : BM1000
Course name : Physio-anatomy-I
Num of credits : 2
Semester : 2
Prerequisites : None

Course Description

This course shall deal with understanding of human anatomy and corresponding physiology of organs & tissue.

Course contents

Course will revolve around Bone/Skeletal, Respiratory, Cardio-vascular and Gastro-intestinal systems.

References

1. Guyton and Hall: Text book of Medical Physiology (2nd SA Edition) ; Ganong: Review of medical physiology (26th Edition) ; BD chaurasia's Human Anatomy (8th edition).

Course code : BM2003
Course name : Intro to Embedded Systems
Num of credits : 3
Semester : 4
Prerequisites : BM2013: Analog and integrated circuits

Course Description

This hands-on course will introduce students to National Instruments chipset, microcontrollers, Raspberry Pi, Arduino.

Course contents

1. Reviewing analog circuits
2. Introducing National Instruments boards/chipset
3. Programming in LabView
4. Introducing Raspberry Pi and Arduino chips
5. Programming Raspberry Pi and Arduino to perform specific tasks

References

1. Agarwal & Lang - Foundations of analog and digital electronic circuits
2. Millman & Halkias - Integrated electronics
3. Malvino & Bates - Electronic principles
4. Gayakwad - Op-Amps & Linear integrated circuits

Course code : BM4021
Course name : Medical image proc Lab
Num of credits : 2
Semester : Odd
Prerequisites : BM4080: Medical Image processing

Course Description

This lab course will introduce students to MATLAB programming with emphasis on developing tools for processing medical images from ultrasound, CT, MRI, etc.

Course contents

1. Introducing MATLAB for image processing
2. Review Fourier transform of images
3. Image filtering and processing
4. Image segmentation
5. Morphological Image processing

References

1. Gonzalez - Digital Image processing using MATLAB
2. Other materials distributed in class

Course code : BM5170
Course name : Ultrasound in Medicine
Num of credits : 3
Semester : Even
Prerequisites : BM3090: Biomedical imaging

Course Description

This course will build upon the physical, mathematical and engineering foundations of medical ultrasound. The course will also provide an overview of recent developments in diagnostic ultrasound, with emphasis in applying theoretical concepts in conceiving novel ideas in diagnostic ultrasound. In addition, this course will introduce students to Therapeutic Ultrasound/Focused Ultrasound/High Intensity FOCused Ultrasound. At the end of the course the student should be able to:

1. Demonstrate a strong understanding of the basic principles of medical ultrasound.
2. Demonstrate a firm understanding of therapeutic ultrasound/FUS/HIFU.
3. Exhibit the ability to understand latest ultrasound literature and develop ideas for novel ultrasound imaging & HIFU.

Course contents

1. Ultrasound and other diagnostic imaging modalities
2. Review of Fourier transform
3. Introduction to image processing and filtering
4. Ultrasound transducers
5. Propagation of ultrasound wave
6. Diagnostic ultrasound imaging modes
7. Imaging systems and applications
8. Bioeffects of ultrasound
9. Review of ultrasound literature
10. Presentations

References

1. Thomas Szabo - Diagnostic Ultrasound Imaging: Inside Out

Course code : BM4091
Course name : Diagnostic Imaging Lab
Num of credits : 3
Semester : Odd
Prerequisites : IDxxxx: Linear systems & SP

Course Description

This course will focus on developing & processing specific tools and resulting images from Ultrasound & Optical imaging modalities.

Course contents

1. Review of image processing
2. Review existing literature in Ultrasound & Optical imaging
3. Proposing novel imaging approaches, and simulating or designing the proposed technique
4. Processing images from real-world images from ultrasound and optical imaging techniques

References

1. Journals and other materials distributed in class

Course code : BM4000
Course name : Regulatory processes and bioethics
Num of credits : 2
Semester : 8
Prerequisites : BM5023: Biomedical Devices

Course Description

This course provides a comprehensive review of medical device regulation in India with an introduction to global regulatory procedures. Regulatory and submission strategy for the Indian markets and Asia Pacific will be a key focus of this course with a blend of practical and applied concepts. Students will also explore important background information regarding the medical devices, ethics and pertaining laws and how select concepts apply to the healthcare system. A thorough review of the registration procedure follows, including discussions on product classification, standards and requirements, and communication CDSCO will be outlines. Other critical areas of discussion include postmarket.

Course contents

Overview of Medical Devices, Revised Medical device rules, Considerations for Developing a Regulatory Strategy , Overview of Medical Device Regulation, and medical device life cycle, medical device standards, quality control and risk analysis

References

1. Val Theisz - Medical Device Regulatory Practices: An International Perspective.
2. Seeram Ramakrishna Lingling Tian Charlene Wang Susan Liao Wee Eong Teo - Medical Devices: Regulations, Standards and Practices.

Course code : BM3011
Course name : Medical biochemistry lab
Num of credits : 2
Semester : Odd
Prerequisites : BM1000; BM1010 : Physio Anatomy I & II

Course Description

This course aims at providing hands on training in different biochemistry laboratory techniques

Course contents

Several qualitative and quantitative analysis of proteins, carbohydrates, urea, uric acid, etc. Blood sugar, blood urea, total cholesterol, serum calcium, serum bilirubin, serum alkaline phosphatase, etc. analysis and estimation.

References

1. Practical textbook of biochemistry for medical students, 3rd Edition, DM Basudevan, Subir kumar Das 2. Manual Of Practical Medical Biochemistry by Jones Evangeline

Course code : BM4190
Course name : BioFabrication
Num of credits : 2
Semester : Odd
Prerequisites : BM5090: Biomaterials-1

Course Description

The aim of this course is to provide insight of 3D bioprinting and allied technologies in biomedical and pharmaceutical applications. It will provide the basics and mechanisms of 3D bioprinting, 3D design software, and 3D tissue/organ printing. In addition, it will also provide nitty gritty of various biofabrication processes, such as the selection and development of biomaterial formulation (bioinks), modulating properties of biomaterials, and controlling different processing conditions. Finally, it will provide state-of-the-art examples of translation of biofabricated products from bench towards the bedside.

Course contents

This course will cover the basics of various 3D bioprinting techniques used in biofabrication; processing of medical imaging data into printable CAD models, and fabricating models on a 3D bioprinter; development of suitable bioinks; critical parameters of bioink for biofabrication; various process parameters and their role in biofabrication; Various 3D bioprinted in vitro, in vivo and ex vivo research models and techniques; in vitro manipulation of cells and biomaterials with a bioprinter to engineer tissues for regenerative medicine or in vitro models; biofabrication-based strategy from bench-to-bed to address a specific clinical problem; ethical issue related to biofabrication.

References

1. Atala et al., Essentials of 3D Biofabrication and Translation. 1st edition, ISBN-13: 978-0128009727.
2. Zhang et al., 3D Bioprinting and Nanotechnology in Tissue Engineering and Regenerative Medicine. 1st edition, ISBN 9780128005477.
3. Forgacs et al., Biofabrication - Micro- and Nano-fabrication, Printing, Patterning and Assemblies, 1st Edition, ISBN 9781455728527

Journal articles

1. Derby B. Printing and prototyping of tissues and scaffolds. Science. 2012. 338:921-6.
2. Seliktar D. Designing cell-compatible hydrogels for biomedical applications. Science. 2012. 336:1124-8.
3. Murphy SV, Atala A. 3D bioprinting of tissues and organs. Nature Biotechnology. 2014. 32:773-85.
4. Pati F, Gantelius J, Svahn HA. 3D Bioprinting of Tissue/Organ Models. Angewandte Chemie International Edition. 2016.55:4650-65.

Course code : BM4011
Course name : Biomicrofab lab
Num of credits : 2
Semester : Even
Prerequisites : BM4190: Biofabrication

Course Description

This course aims to provide hands on training in different biofabrication and microfabrication technologies used in Biomedical Engg.

Course contents

Different fabrication techniques for biomaterials and their basic characterization. Different biomaterials such as nanofibers, 3D scaffold, 3D printed device etc for different applications

References

1. Atala et al., Essentials of 3D Biofabrication and Translation. 1st edition, ISBN-13: 978-0128009727.
2. Zhang et al., 3D Bioprinting and Nanotechnology in Tissue Engineering and Regenerative Medicine. 1st edition, ISBN 9780128005477.
3. Forgacs et al., Biofabrication - Micro- and Nano-fabrication, Printing, Patterning and Assemblies, 1st Edition, ISBN 9781455728527

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1. Derby B. Printing and prototyping of tissues and scaffolds. Science. 2012. 338:921-6.
2. Seliktar D. Designing cell-compatible hydrogels for biomedical applications. Science. 2012. 336:1124-8.
3. Murphy SV, Atala A. 3D bioprinting of tissues and organs. Nature Biotechnology. 2014. 32:773-85.
4. Pati F, Gantelius J, Svahn HA. 3D Bioprinting of Tissue/Organ Models. Angewandte Chemie International Edition. 2016.55:4650-65.

#N/A

Course code : BM5090
Course name : Biomaterials-1
Num of credits : 2
Semester : 5
Prerequisites : BM1000; BM1010 : Physio Anatomy I & II

Course Description

This is an introductory course of biomaterials where materials are using for human health care. This course will provide a broad overview to the student about different types of materials and their application in human health care.

Course contents

Introduction of different generations of biomaterials, Different class of materials used in medicine, Understanding the general properties of different classes of materials to be used as biomaterials, Familiarity with biological response to biomaterials, biocompatibility and hemocompatibility.

References

1. An Introduction to Materials in Medicine, Edited by Buddy D. Ratner, Allan S. Hoffman, Frederick J. Schoen, Jack E. (Elsevier Academic Press, ISBN: 0-12-582463-7)
2. Biomaterials - Temenoff and Mikos (Pearson Prentice Hall; ISBN 0-13-009710-1).

Course code : BM5141
Course name : Biomaterials-2
Num of credits : 2
Semester : Even
Prerequisites : BM5090: Biomaterials- 1

Course Description

This is an advance course of biomaterials where materials are using for human health care. In this course student will know about biomaterials, desing and as well as fabricate (partly) of biomaterials device for particular application.

Course contents

Advance biomaterials for heat, bone, cartilage, skin, dental, ocular applications

References

1. An Introduction to Materials in Medicine, Edited by Buddy D. Ratner,
2. Scientific Journals

Course code : BM3030
Course name : Medical nanotech-I
Num of credits : 2
Semester : 4
Prerequisites : BM1000; BM1010 : Physio Anatomy I & II

Course Description

This is a highly interdisciplinary course for students who are interested in learning about the emerging field of nanoscience and nanotechnology and its application in biology and medicine. To capture the excitement of this emerging field, in this course student will be familiarized with fundamentals of nanoscience and Nano-scale engineering, and their potential application in the human health care system.

Course contents

This course will emphasize emerging nanotechnologies and its biomedical applications including fundamental of nanomaterials and nanoengineering, nanotoxicology, nanotechnology for drug delivery, regenerative medicine, imaging, and diagnostic system and translating nano-medicines into

References

1. Jain, Kewal K. The Handbook of Nanomedicine. Humana Press, 2008.
2. Nanomedicine Design of Particles, Sensors, Motors, Implants, Robots, and Devices. Ed. Mark J. Schulz, Vesselin N. Shanov, and Yeoheung Yun. Artech House, 2009. ISBN-13: 978-1-5969-3279-1
3. Nanotechnology: Volume 5: Nanomedicine. Ed. Viola Vogel. Weinheim: Wiley-VCH Verlag GmbH & Co. KGaA, 2009. ISBN-13: 978-3-5273-1736-3

Course code : BM6080
Course name : Advanced Biomechanics
Num of credits : 2
Semester : Even
Prerequisites : IDxxxx: Engineering Mechanics, BM3040: Introduction to Biomechanics

Course Description

The course aims at providing an advanced overview of non-linear soft tissue mechanics, tissue growth, biological fluid flows and solid and fluid interactions in the biological systems

Course contents

Soft tissue mechanics, Fung's and Holzapfel models, Artery blood flow modeling, transport in GI tract, Experiments on soft polymer/tissue characterization

References

1. An introduction to Biomechanics, Solids, Fluids, Analysis and Design
2. Nonlinear elasticity book by Ogden
3. Other materials distributed in class

Course code : BM4040
Course name : Mechano-biology
Num of credits : 3
Semester : Odd
Prerequisites : BM3040: Introduction to Biomechanics

Course Description

Introduction and overview of the role of physical forces and flows on/in cells and macromolecules in determining biological dynamics

Course contents

Force generation at the cellular scales, role of non-equilibrium statistical mechanics, stretch activated growth factor receptors, myosin motors, cytoskeletal elements, ECM, and other molecular structures and signalling molecules in mechanotransduction. Relevant disease models.

References

1. Mechanobiology: from Molecular sensing to Disease, Glen. L. Neibur
2. Other materials distributed in class

Course code : BM6070
Course name : Biomicrofluidics
Num of credits : 3
Semester : Even
Prerequisites : IDxxxx: Engineering Mechanics, BM3040: Introduction to Biomechanics

Course Description

Course is focused on providing a detailed understanding on the design and fabrication of chip scale analytical devices for biological and biomedical applications

Course contents

Lab on Chip Microfluidic theory, design, biological and biomedical applications

References

1. Introduction to Microfluidics (Patrick Tabeling)
2. Other materials distributed in class

Course code : BM3001
Course name : Biomechanics Lab
Num of credits : 2
Semester : Even
Prerequisites : BM3040: Introduction to Biomechanics

Course Description

The aim of this course is to provide hands on training in various biomechanical characterization techniques used in biomedical engg.

Course contents

Mechanical characterization of biomaterials and tissues like tensile, compression, bending, fatigue, creep, stress relaxation,

References

1. An introduction to Biomechanics, Solids, Fluids, Analysis and Design
2. Relevant Journal publications

Course code : BM3040
Course name : Introduction to Biomechanics
Num of credits : 2
Semester : 5
Prerequisites : BM1000; BM1010 : Physio Anatomy I & II

Course Description

The course aims at providing an introduction to constitutive models of linear isotropic, anisotropic and viscoelastic behavior of biomechanical elements

Course contents

Linear elastic and viscoelastic models, biological and biomedical applications

References

1. An introduction to Biomechanics, Solids, Fluids, Analysis and Design
2. Relevant Journal publications

Course code : BM2013
Course name : Analog & Integrated circuits
Num of credits : 3
Semester : 3
Prerequisites : IDxxxx: Basics of Elec Engg

Course Description

The course provides a theoretical basis of advanced analog and integrated circuits, especially operational amplifier implementations for applications. It also introduces safe handling practices of electronic lab equipment to students.

Course contents

Safe use of bread-board, oscilloscope, power supplies for testing operations. Bias voltage supplies; Diode: half-wave, full-wave rectifiers; BJT & JFET amplifier characteristics; Operational amplifiers - mathematical operations, active filter implementations - single & two-pole, Sallen-Key; Differential amplifier & Instrumentation amplifier implementations.

References

1. Agarwal & Lang - Foundations of analog and digital electronic circuits
2. Millman & Halkias - Integrated electronics
3. Malvino & Bates - Electronic principles
4. Gayakwad - Op-Amps & Linear integrated circuits

Course code : BM5023
Course name : Biomedical devices
Num of credits : 2
Semester : 7
Prerequisites : BM5013: Sensors and transducers in healthcare

Course Description

This course is intended to impart understanding of various biosignals, explain their origin, sensing, signal conditioning and meaningful representation. Temperature, physiological pressures and flows, volume sensing, bioelectricity from muscles and eyes, the related biological systems and their physiology will be discussed. The course includes work involving advanced linear integrated circuit design, interfacing and real-time processing of biosignals and understanding the diagnostic value in biosignals will be discussed.

Course contents

Temperature sensing and signal conditioning, Pressure sensing - basics of cardio-pulmonary system, physiological pressures and sounds, invasive and non-invasive measurements. Flow and volume sensing in pulmonary function - sensing of the mechanics of breathing, spirometer interface with LabVIEW, computation of volumes and capacities during a pulmonary function test, mechanical ventilation, applications. Bioelectric sensing: Electro-oculogram & Electro-retinogram: Biopotentials of the eye, instrumentation, instrumentation, diagnostics. Electromyography - Origin, processing, detection of muscle fatigue, nerve conduction velocity test. Real-time ECG monitor with heart rate detection.

References

1. John G. Webster-Medical Instrumentation Application and Design, 4th Edition (2009)
2. Leif Sornmo, Pablo Laguna Bioelectrical Signal Processing in Cardiac and Neurological Applications Biomedical Engineering

Course code : BM5080
Course name : Neurotechnology & BCI theory
Num of credits : 2
Semester : Even
Prerequisites : IDxxxx: Linear systems & SP, BM3000: Foundations of Natural intelligence

Course Description

This course will introduce neurotechnology, especially neuroimaging, neuromodulation and Brain machine interfaces (BMI). Neuroimaging encompasses use of electrophysiology techniques like EEG and fusion of functional activity with anatomical information. Neuromodulation is the large variety of techniques used to assist, treat injuries to various parts of the nervous system. BCI on the other hand is a channel of communication between a brain and a machine (typically a computer). These techniques have a variety of medical as well as computer applications. Students are also expected to gain insight into circuitry involved in movement in healthy and disorder cohorts such as Parkinsons disease.

Course contents

The neuroscientific concepts that help entrain subjects to learn from abstract cues and rewards will be discussed. Mathematical models and stochastic processes to capture the learning processes will be discussed. Various modalities to record the subject's intent such as EEG, MEG, fNIRS, EMG will be discussed. Furthermore, lab component will include design of BMI system with real-time processing of electrophysiological signals along with feedback. Students are expected to gain end-to-end knowledge of the field of neuroimaging, neuromodulation, BMI and how signal processing and machine learning concepts can be applied to build better BCI systems.

References

1. Analyzing Neural Time Series Data_ Theory and Practice-Mike X Cohen-The MIT Press (2014)
2. Robert Coben, James Evans-Neurofeedback and Neuromodulation techniques and applications, 1st edition (2011)
3. Niedermeyer & Lopez da Silva - Niedermeyer's Electroencephalography: Basic Principles, Clinical Applications, and Related Fields (Lippincott Williams and Wilkins) 5th Edition. ISBN: 9780781789424
4. Malmivuo & Plonsey – Bioelectromagnetism- Principles and Applications of Bioelectric and Biomagnetic Fields (Oxford University Press) 1995

Course code : BM5081
Course name : Neurotechnology & BCI lab
Num of credits : 2
Semester : Even
Prerequisites : BM4070: Neurotechnology & BCI theory

Course Description

This course is meant to provide practical insight into using neurotechnology, especially EEG and other electrophysiology tools to implement real-time processing algorithms, machine learning techniques to implement a lean brain-computer interface. Students are expected to gain end-to-end knowledge of the field of neuroimaging, neuromodulation, BMI and how signal processing and machine learning concepts can be applied to build better BCI systems.

Course contents

EEG, fNIRS and EMG will be used to implement BCI and discussed. Furthermore, lab component will include design of BMI system with real-time processing of electrophysiological signals along with feedback. Students are expected to implement basic acquisition, real-time processing and establish an interface with openBCI framework and perform experiments of brain control of external devices.

References

1. Analyzing Neural Time Series Data_ Theory and Practice-Mike X Cohen-The MIT Press (2014)
2. Robert Coben, James Evans-Neurofeedback and Neuromodulation techniques and applications, 1st edition (2011)
3. Niedermeyer & Lopez da Silva - Niedermeyer's Electroencephalography: Basic Principles, Clinical Applications, and Related Fields (Lippincott Williams and Wilkins) 5th Edition. ISBN: 9780781789424
4. Malmivuo & Plonsey – Bioelectromagnetism- Principles and Applications of Bioelectric and Biomagnetic Fields (Oxford University Press) 1995

Course code : BM2033
Course name : Probability & Random Processes
Num of credits : 2
Semester : 4
Prerequisites : MA2110:Probability and Statistics

Course Description

This is an undergraduate's introduction to mathematical treatment of uncertainty. It introduces students to basics of Probability and random variables and random processes. It also introduces students to some common tools based on these concepts that are widely used, such as Estimation and MCMC methods. Apart from theory tutorials, it will cover Lab tutorials, assignments and a short lab project

Course contents

Random variables

- Probability distributions
- Discrete Random Variables, Continuous & General Random Variables
- Random Vectors
- Function of Random Variables
- Expectation, Variance, Conditional Expectation
- Bounds: Jensen, Markov, Chebyshev, Chernoff
- Law of large numbers, Central limit Theorem
- Bayes Rule

Random Processes

- Discrete Time Markov Chains
- Poisson Process
- Markov Decision Process
- Neyman-Pearson Theorem
- Estimation
- LLSE, MMSE
- Cramer-Rao bound
- Kalman Filter

Algorithms

- Viterbi Algorithm
- Expectation Maximization & Clustering
- PCA
- Markov chain Monte Carlo methods

References

1. Bertsekas, D. P., & Tsitsiklis, J. N. (2002). Introduction to Probability
2. Walrand, J., 2004. Lecture notes on probability theory and random processes. University of California, Berkeley, p.193.

Course code : BM2043
Course name : Algorithms and data structures Lab
Num of credits : 2
Semester : 3
Prerequisites : ID2230: Elec Engg

Course Description

This course provides an introduction to algorithms. It also includes lab exercises for implementation of data structures and algorithms on them

Course contents

Theory:

- Complexity of algorithms
- Greedy algorithms
- Dynamic programming
- Shortest Paths
- Choice of data structures and algorithms

Lab:

- Implementation of Linked lists, Stacks, Trees, Graphs
- Searching, sorting and traversals
- Experimental verification of complexities
- Implementations of Greedy and Dynamic programming techniques
- Short Project: System implementation using appropriate data structure, operations and algorithms

References

1. Horowitz, E. and Sahni, S., Fundamentals of computer algorithms. 1978. Pitman, London.

Course code : BM6140
Course name : Theoretical & computational neuroscience
Num of credits : 2
Semester : Even
Prerequisites : BM3000: Natural Intelligence, Foundations of

Course Description

This course provides an introduction to mathematical modeling of elements in the nervous system viz ion channels, neurons, synapses, networks and learning therein.

Course contents

1. Genesis of electrical activity in cells, resting membrane potentials
2. Neuron equivalent circuits and passive propagation in neurons
3. Hodgkin-Huxley equations and conductance based models
4. Ion channels and their diversity
5. Simple neuron models and analysis using dynamical systems concepts
6. Chemical and electrical synapses and their models
7. Neuronal networks and techniques for mathematical analysis
8. Models of learning and memory in the neuron and the network
9. Neural coding (Rate, temporal, population)

References

1. Ermentrout, G.B. and Terman, D.H., 2010. Mathematical foundations of neuroscience (Vol. 35). Springer Science & Business Media.
2. Johnston, D. and Wu, S.M.S., 1994. Foundations of cellular neurophysiology. MIT press.

Course code : BM4025
Course name : Natural Intelligence, Implementations of
Num of credits : 2
Semester : Odd
Prerequisites : BM3000: Natural Intelligence, Foundations of

Course Description

This course involves one or more short projects centred around implementations of landmark natural intelligence papers

Course contents

The course involves creating software or hardware implementations of popular and landmark biomimetic formulations in the field of human/computer vision, human/robotic movement, human/artificial decision making

References

1. Purves, D.E., Augustine, G.J., Fitzpatrick, D.E. and Katz, L.C., 1997. Neuroscience. Sunderland, MA, US: Sinauer Associates.
2. Gazzinga - The Cognitive Neurosciences
3. Material distributed in class

Course code : BM4061
Course name : Movement sciences lab
Num of credits : 2
Semester : Odd
Prerequisites : BM4060: Movement science

Course Description

This course gives a hands on experience of tools, techniques and equipments for measurement of movement. On completion of the course, a student will be able to understand and perform EMG, Motion capture using optical and electromagnetic sensors, processing like inverse kinematics using popular simulators such as OpenSim and create neuromechanical systems using the NEUROiD platform. It is a recommended course for senior undergraduate and graduate students of biomedical, electrical, mechanical and computer sciences and engineering with an interest in assistive/rehab technologies, movement disorders or robotics.

Course contents

- Electromyography
 - Passive and active electrodes and usage
 - Identifying muscles
 - Maximum voluntary contraction experiments for various muscles
 - Protocols for noise free acquisition
 - Creating pipelines for data acquisition and analysis
- Optical Motion capture
 - Setting up the MoCap systems
 - Marker Sets and Design
 - Deploying markers
 - Calibration and acquisition protocols
- Electromagnetic Motion capture
 - Setup and precautions
 - Calibration and acquisition protocols
- Force Plates
 - Setup
 - Calibration and acquisition protocols
- Gaze tracking
 - Synchronization methods
 - Data processing
- Choosing a model in Open-Sim
- Forward and inverse modelling
- Data compaction, denoising and visualization
- Creating virtual neuro-mechanical systems using NEUROiD
- Neural controller design
- Musculoskeletal model design
- Recreating movement

References

1. Iyengar, R. S., Pithapuram, M. V., & Singh, A. K. (2019). Curated Model Development Using NEUROiD: A Web-Based NEUROMotor Integration and Design Platform. 13(August), 1–30.
2. Millard M, Uchida T, Seth A, & Delp S L. (2013). Flexing computational muscle: Modeling and simulation of musculotendon dynamics. Journal of

Course code : BM4081
Course name : Computational neuroscience lab
Num of credits : 2
Semester : Odd
Prerequisites : BM6140: Theoretical and Computational Neuroscience

Course Description

This course gives a hands on experience of computational modeling for Neurosciences. It largely utilizes the popular NEURON simulator with a Python interface. It also familiarizes students with other available simulators and computational tools available for network modeling. It is a recommended course for senior undergraduate and graduate students of biomedical, electrical and computer sciences and engineering with an interest in neuromorphic and bio-inspired systems.

Course contents

- Introduction to NEURON
- mod, hoc and Python interpreters in NEURON
- modeling passive properties
- modeling Hodgkin-Huxley neurons
- modeling Spatial extent of neurons
- modeling cell-groups, synapses and networks of neurons
- implementing markovian models of channel states
- Simple neuron models using generic ODE solvers
 - Fitzhugh-Nagumo
 - Izhikevich
 - Morris-Lecar
- Dynamical systems analysis of neurons and networks with Matlab
- Network modeling using BRIAN
- Integrate and fire neuron models
- Spiking neuron models; Py-NEST
- Creating neural systems by curated integration - NEUROiD
- Integrating physiology and anatomy
- modeling spike statistics:
 - inter spike interval and exponential distributions
 - Poisson spike trains
- Concept to model exercises

References

1. Hines, M.L. and Carnevale, N.T., 1997. The NEURON simulation environment. Neural computation, 9(6), pp.1179-1209.
2. Iyengar, R.S., Pithapuram, M.V., Singh, A.K. and Raghavan, M., 2019. Curated model development using NEUROiD: A web-based NEUROmotor integration and Design platform. Frontiers in neuroinformatics, 13, p.56.

Course code : BM4060
Course name : Movement science
Num of credits : 2
Semester : Even
Prerequisites : BM3000: Natural Intelligence, Foundations of

Course Description

This course provides an understanding of human movement and its mechanistic basis. Covering a range of topics from biomechanics and neuromotor control, the course hopes to impart to senior undergraduate and graduate students an intuitive and mathematical understanding of the various aspects of human movement.

Course contents

- Physics of the Musculoskeletal system
- Anatomy of the musculoskeletal system
- Quantitative Muscular physiology
- Spinal circuits of movement
- Cortical Control of movement
- Modulation of movement
- Measurement in NeuroMechanics

References

1. Herman, I.P., 2016. Physics of the human body. Springer.
2. Silverthorne, D.U., 2004. Human physiology. Pearson.
3. Pierrot-Deseilligny, E. and Burke, D., 2012. The circuitry of the human spinal cord: spinal and corticospinal.
4. Purves, D.E., Augustine, G.J., Fitzpatrick, D.E. and Katz, L.C., 1997. Neuroscience. Sunderland, MA, US: Sinauer Associates.
5. Winter, D.A., 2009. Biomechanics and motor control of human movement. John Wiley & Sons
6. Enoka, R.M., 2008. Neuromechanics of human movement. Human kinetics.
7. Nordin, M. and Frankel, V.H. eds., 2001. Basic biomechanics of the musculoskeletal system. Lippincott Williams & Wilkins.

Course code : BM3000
Course name : Natural intelligence, Foundations of
Num of credits : 3
Semester : 5
Prerequisites : BM2033: Probability and Random processes; BM1000; BM1010 : Physio Anatomy I & II

Course Description

This course provides an introduction to the principles of natural intelligence that provided the inspiration for Artificial Intelligence. This course will deal with natural counterparts of several AI and ML concepts that are widespread today such as the perceptron, neural nets and CNNs. This course is recommended for senior undergraduate and graduate students of Biomedical, Electrical and Computer Science Engineering with interest in development of biomimetic AI and ML approaches

Course contents

- Motivation for NI-AI: comparison of the natural and artificial intelligence machines
- Levels of abstraction in study of brain and intelligence
 - ion channel, neuron, networks, systems
- Primer to the working of brain
 - Visual, auditory and somatosensory systems
 - Motor systems, speech and language
 - Memory and learning
- Natural counterpart of the primordial perceptron
 - Non-linear integration of inputs
 - Voltage gated ion channels and the perceptron activation functions
 - Coincidence detection
 - Natural switches to tweak the perceptron behaviour
- Natural counterpart of AI's neural nets and learning through backprop
 - Learning and plasticity in synapses
 - Hebbian learning
 - Spike timing dependant plasticity
 - Stability of learning and BCM rule
- The language of neurons: Neural coding
 - Spike codes: frequency, temporal, population codes
 - Information encoding and decoding
 - Bayesian Brain
- Visual system and the convolutional neural networks
- Reward behavior and reinforcement learning:
 - Dopamine-based reward prediction errors vs difference learning, distributed reinforcement learning
- Movement disorders circuits and models:
 - PD - BG-cortical circuits, modelling disease
 - GEN models (Chakravarthy et al., 2017)
- Body image vs Bayesian inference- disease models - anosognosia, phantom limb - Ramachandran's mirror box

References

1. Cooper, L.N., 2004. Theory of cortical plasticity. World Scientific.
2. Rieke, F., Warland, D., Van Steveninck, R.D.R. and Bialek, W.S., 1999. Spikes: exploring the neural code (Vol. 7, No. 1). Cambridge: MIT press.
3. Doya K, Ishii S, Pouget A and Rao R P eds. 2007. Bayesian brain: Probabilistic approaches to neural coding. MIT press

Course code : BM3090/BM6090
Course name : Biomedical imaging
Num of credits : 3
Semester : 4
Prerequisites : BM1000; BM1010 : Physio Anatomy I & II

Course Description

This course provides a broad overview of different imaging modalities, their ability to image the human body, and their clinical applications.

Course contents

Introduction to Xrays, Tomography, CT,PET, SPECT, Introduction to Ultrasound imaging, MRI, Optical imaging modalities

References

1. Medical Instrumentation Application and Design, John Webster Ed. John Wiley & Sons 2009.

Course code : BM5013
Course name : Sensors & transducers in healthcare
Num of credits : 2
Semester : 5
Prerequisites : BM2013: Analog and integrated circuits, BM2003: Introduction to Embedded systems

Course Description

This course is intended to understand the origin of signals in biosystems and living organisms, their sensing, detection and meaningful processing for practical diagnostic sensing applications. Various engineering aspects of the detection, acquisition, processing, and display of signals, biomedical sensors for measurements of biopotentials such as ECG, displacement sensors, gas analyzers will be addressed in this course. The course includes work involving basic electronics, sensor design and interfaces for building complete biomedical instrumentation.

Course contents

Generalized design of bioinstruments, noise suppression techniques, Bioinstrument characteristics – basics of regulatory processes. Amplifiers & filters. Sensors & signal conditioning circuits for displacement sensing, gas analyzers, pulse oximetry. Principles bioelectric transduction, ECG-origin, physiology, 12-lead system, instrumentation, rhythms, arrhythmias, pathology, diagnostics. Simulators for pacemaker, defibrillator, heart rate detection.

References

1. John G. Webster-Medical Instrumentation Application and Design, 4th Edition (2009)
2. Leif Sornmo, Pablo Laguna Bioelectrical Signal Processing in Cardiac and Neurological Applications Biomedical Engineering

Course code : BM6246
Course name : Clinical immersion & Biodesign
Num of credits : 2
Semester : 8
Prerequisites : BM1030: Bioengineering, BM5023: Biomedical devices

Course Description

This course focuses on a design process thinking for Innovations in healthcare. The course will begin with a clinical immersion component to obtain unmet needs in the healthcare setting. The students will then undergo training in the biodesign process thinking.

Course contents

Students will visit some hospitals and observe patient experiences and interactions with medical devices. They will also interact with physicians to better understand their perspectives. The curriculum aims at exposing the students to unmet needs in the clinic and introduce a systematic design process thinking to arrive a solution to the problems. The course will have detailed case studies and problem-based learning sessions on Biodesign. A short report is required to be submitted as a course evaluation, which is aimed to encourage students to think innovatively about biomedical engineering.

References

1. Zenos, Makover, P. Yock, Biodesign, Cambridge University Press (2009)
2. Resources distributed in class

Course code : BM4020
Course name : Biophotonics
Num of credits : 3
Semester : Odd
Prerequisites : BM3090: Biomedical Imaging

Course Description

This course is intended to introduce the concepts of biophotonics and interaction of light with biological matter. Biophotonics play a leading role in modern diagnostics, imaging and therapy. This course introduces principles of various biomedical optical imaging devices and techniques, lasers for biomedical applications, light-activated and light-guided therapy and biophotonic devices. This course is offered to newcomers and researchers interested in biophotonics and nanobiophotonics. This is offered as a core for Phd students in dept of Biotechnology/Biomedical Eng as well as an elective for Btech/Mtech students interested in this area.

Course contents

Biophotonics Topic introduction, Multidisciplinary nature, and role of photonics in biomedical engineering.
Fundamentals of Light and Matter , Nature of Light, Dual Character of Light, Propagation of Light as Waves, Coherence of Light, Light as Photon Particles, Different Light Sources, lasers and Introductory Concepts. Lasers: A New Light Source, Principles of Laser action , Classification of Lasers, Some Important Lasers for Biophotonics. Current Laser Technologies, Light-Matter Interactions, absorption, emission, scattering. Raman, Nonlinear Processes.
Basics of Biology, Introductory Concepts, Cellular Structure, Various Types of Cells, Cellular Processes, Organization of Cells into Tissues, Types of Tissues and Their Functions , Tumors and Cancers.
Photobiology-the Core of Biophotonics, Interaction of Light with Cells, Light Absorption in Cells , Light-Induced Cellular Processes, Photochemistry Induced by Exogenous Photosensitizers, The Human Eye and Vision, Pulse Oximetry.

References

1. P. Prasad, Introduction to Biophotonics, John Wiley (2003).
2. J. W. Goodman, Introduction to Fourier optics, 2nd ed. The McGraw Hill company, (2006).

Course code : BM4080
Course name : Medical image proc & analysis
Num of credits : 2
Semester : Odd
Prerequisites : IDxxxx: Linear systems & SP

Course Description

This course will cover topics in Image Processing, enabling students to extend their knowledge on Signals & Systems, and DSP. In addition, this course will expose students to various image processing tools, and on image obtained using different modalities (Ultrasound, MRI, CT, etc.)

Course contents

2D Fourier of Images, Morphological Image processing, Image segmentation

References

1. Digital Image Processing (4th Edition), Rafael C. Gonzalez & Richard Woods

Course code : BM2023
Course name : Basic Bioinformatics
Num of credits : 2
Semester : 4
Prerequisites : BM1000; BM1010 : Physio Anatomy I & II

Course Description

Practical course to learn diseases at genetic and proteomic level.

Course contents

Biological Data from different sources in internet. Sequence of DNA, protein, and finding protein structures based on the data. Formats of databases. Conventions for database indexing and specification of search terms, Common sequence file formats. Organism specific databases and analysis. Sequence Similarity Searches: Local versus global. Distance metrics. Methods of sequence alignment, FASTA, BLAST and PSI BLAST. Multiple Sequence Alignment and software tools for pairwise and multiple sequence alignment. Phylogenetics structural bioinformatics genomics and proteomics.

References

1. Bioinformatics and Functional Genomics (2nd edition) by Jonathan Pevsner, Wiley-Blackwell.
2. Essential Bioinformatics. Jin Xiong, 2007.

Course code : BM3023/BM6023
Course name : Cell Technology
Num of credits : 3
Semester : 5
Prerequisites : BM1000; BM1010 : Physio Anatomy I & II

Course Description

Cell and molecular biology techniques.

Course contents

This course is intended for practical handling experience for students for culture of mammalian cells. They should learn detailed step-wise protocols in culturing, freezing, splitting of mammalian cells. In addition, they should be able to see the cells found in blood. In addition, they will be taught molecular biology techniques such as DNA isolation, RNA isolation, polymerase chain reaction, and transfection. They should also learn basic molecular biology methods and their interpretation with respect to cell culturing techniques.

References

1. Molecular Cell Biology, 2012 by Harvey Lodish and Arnold Berk.
2. Culture of animal cells. Editor: R. Ian Freshney. 6th edition.

Course code : BM1010
Course name : Physio-anatomy-II
Num of credits : 2
Semester : 3
Prerequisites : BM1000: Physio Anatomy - I

Course Description

Physiology and anatomy of respiratory system, blood vessels, Gastro-intestinal tract, central nervous system.

Course contents

This course is intended for basic understanding of human physiology in the engineers' perspective. The students need to understand the cellular and physiological systems; and their dysfunction by applying engineering and mathematics knowledge. The important systems which may be covered are nervous system, cardiovascular, respiratory, smooth muscles, gastro-intestinal tract.

References

1. Ganong's textbook of physiology.

Course code : BM4120
Course name : Tissue engg & Regenerative med
Num of credits : 3
Semester : Even
Prerequisites : BM4090: Biomaterials- 1

Course Description

Engineering of any biological tissue using cells, biomaterials and bioreactor. How to test them by different methods.

Course contents

Tissue engineering: fundamentals and current status; Stem cells: embryonic and mesenchymal stem cells; cell differentiation; Extra-cellular matrix components and their regulation of cell behavior; In vitro and in vivo testing of biomaterials. Bioreactor; Cell migration; Growth factors; Different approaches for angiogenesis and its importance.

References

1. Principles of Tissue Engineering. Robert Lanza, Robert Langer, Joseph P. Vacanti. Academic press.

Course code : BM2053
Course name : Mathematical models & systems biology
Num of credits : 2
Semester : 3
Prerequisites : BM1000; BM1010 : Physio Anatomy I & II

Course Description

The aim of this course is to introduce basic principles of the systems biology (experimental as well as computational) and general approaches towards system level mathematical modeling of biological systems.

Course contents

Introduction to systems biology from experimental as well as mathematical view point. Tools for the analysis (time and frequency space) of the experimental data, mathematical modeling from individual molecules to signaling pathways.

References

1. Thinking in systems: a primer
2. Introduction to Systems biology by Uri Alon

Course code : BM2000
Course name : Control Systems
Num of credits : 1
Semester : 4
Prerequisites : None

Course Description

This course will provide understanding of the fundamentals of control systems starting from basics of control theory to some examples of biomedical engineering applications. This course is for all BME undergraduates and will be essential for the understanding of higher level courses for almost all BME specializations.

Course contents

Introduction to control, transfer function, state-space model, stability analysis and criteria, P, PI, PD, and PID controllers; examples from biomedical engineering such as blood glucose regulation, artificial ventilation, cardiac assist devices, drug delivery systems etc.

References

1. Control Systems Engineering by Norman Nise
2. Automatic Control Systems in Biomedical Engineering by Cañete et al.

Course code : BM4051
Course name : Computational Biomechanics Lab
Num of credits : 2
Semester : Even
Prerequisites : BM3040: Introduction to Biomechanics

Course Description

This course seeks to give students hands on training on the computational and application aspects of the biomechanics covered in the theory courses. In the course students will be exposed to the numerical simulation techniques/relevant softwares for the analysis of solid and fluid biomechanics in the form of several lab sessions.

Course contents

1. Overview of the computational methods used in biomechanics
2. Computational analyses of bone and soft tissue mechanics using abaqus/comsol/FENICS etc.
3. Blood and peristaltic flow modeling using Fluent/comsol/FENICS etc.
4. Computational modeling of fluid structure interaction in biological systems.

References

1. JN Reddy: An Introduction to FEM
2. Niyogi et al.: Introduction to CFD

Course code : BM5160
Course name : Mechanics of bio-fluids
Num of credits : 2
Semester : Even
Prerequisites : BM3040: Introduction to Biomechanics

Course Description

This course is intended to cover advanced topics on the mechanics of biological fluids. The focus of this course is going to be on the bio-mechanical aspects of the biological flows such as oscillatory blood flows (inertial as well as non-inertial), peristaltic flows in gastrointestinal tracts, non-Newtonian rheology of bio-fluids, mechanics of active bio-fluids, and effect of organ geometries on the biological flows.

Course contents

This course will start with the basics of fluid mechanics and will cover specific examples from the biological systems. The specific topics include oscillatory flows, peristaltic flows, active biological fluids, fluid-structure interactions, emergence of non-Newtonian nature of bio-fluids due to their deformable components with application in microfluidics.

References

1. Introduction to Fluid Mechanics by Batchelor
2. Low Reynolds Number Hydrodynamics by Happel and Brenner
3. Relevant research papers